Temporal Generalizability and Predictive Stability in Municipal Finance: Parallel Evidence from Norway and the United States

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Abstract

This study examines whether municipal financial characteristics remain associated with subsequent fiscal and service outcomes across time and national settings, and whether future municipal debt burden can be forecast stably out of time. The inferential program begins with Norwegian KOSTRA data, applies held-out temporal validation and an official municipality-continuity correction, then independently constructs a U.S. design from Census government-finance data. Cross-country comparison is restricted to conceptually harmonizable constructs rather than accounting-identical measures. A separate predictive program uses rolling out-of-time evaluation with a forecast-origin embargo so that no training outcome postdates the information available at the forecast origin.

Three Norway associations passed discovery false-discovery-rate control. After official continuity correction, formal discovery-versus-validation shift tests supported change for unrestricted revenue per capita versus accumulated balance (BH q=0.0058) and self-investment financing versus debt burden (BH q=0.0165), while the home-service association showed no supported period shift (BH q=0.732). The debt validation estimate remained approximately zero and is not interpreted as evidence of a positive effect. In the United States, the headline two- and three-year interest-burden coefficients were dominated by influential observations and denominator tails. Trimming, winsorization, common-government restriction, and a fixed $1 million revenue floor sharply reduced the discovery magnitudes; only the two-year interest shift persisted across several alternative support rules, whereas the three-year temporal change did not. No statistical Norway-U.S. replication was established.

Forecasting also differed by country. Norway showed strong state-based predictability relative to a pooled historical mean, although Persistence was not clearly superior to pooled AR(1) in paired-fold comparisons. In the United States, RMSE-based selections were weakly separated and the largest 1% of row-level errors contributed about 99.3-99.9% of squared error on common support. All-model MAE rankings disagreed with RMSE rankings at the two- and three-year horizons and changed again under the fixed revenue floor. The results support cautious claims about temporal generalizability and predictive stability rather than universal fiscal relationships or a common cross-country forecasting architecture.

1. Introduction

Measures of local fiscal condition are used to describe vulnerability, debt capacity, service pressure, and the risk of fiscal distress. A long literature has proposed financial-condition indicators and models of local fiscal stress, including work on fiscal health, composite distress prediction, and the determinants of local-government debt (Hendrick 2004; Kloha, Weissert, and Kleine 2005; Wang, Dennis, and Tu 2007; Gorina, Maher, and Joffe 2018; Balaguer-Coll, Prior, and Tortosa-Ausina 2016; Ehalaiye, Botica-Redmayne, and Laswad 2017). Yet a coefficient or model that is convincing in one period, specification, or institutional system need not remain stable when the time period, sample composition, accounting regime, or national context changes.

This study treats that instability as an empirical object rather than as a nuisance to be hidden. The terminology follows the distinction between computational reproducibility, replication, and broader generalizability emphasized in the reproducibility literature (Goodman, Fanelli, and Ioannidis 2016; National Academies of Sciences, Engineering, and Medicine 2019; Munafò et al. 2017). The main concern here is *temporal generalizability*: whether an association or predictive pattern obtained in one period remains directionally and substantively similar in later periods. Statistical significance in one period and non-significance in another is not itself a formal test of change (Gelman and Stern 2006), so the revised analysis also compares coefficients directly.

The program contains two deliberately separate evidence layers. The inferential layer asks whether specified financial characteristics are associated with later fiscal or service outcomes after fixed effects and controls. The predictive layer asks how well future debt burden can be forecast from information available at the forecast origin. Predictive

performance is not used to reopen inferential findings, inferential significance is not used for predictive feature selection, and neither layer is interpreted causally.

The paper makes four contributions. First, it combines discovery, held-out temporal validation, formal coefficient-shift tests, and municipality-identity correction in a longitudinal municipal-finance design. Second, it shows that the striking U.S. interest-burden result is primarily an influence- and denominator-tail problem before it is a temporal-generalizability problem. Third, it demonstrates the importance of forecast-origin timing in multi-horizon municipal prediction. Fourth, it places parallel Norway and U.S. evidence side by side as a secondary comparison without treating conceptually similar fiscal measures as accounting-identical quantities.

1.1 Research questions

RQ1. Which Norwegian municipal financial characteristics are associated with subsequent fiscal or service outcomes, and how do those associations change in held-out years after official municipality-continuity correction?

RQ2. Do analogous U.S. municipal-finance associations remain stable under influence diagnostics, regime windows, and formal discovery-versus-validation coefficient-shift tests?

RQ3 (secondary cross-country question). Among the small subset of constructs that are conceptually harmonizable across Norway and the United States, is there evidence of statistical replication or only limited descriptive directional similarity?

RQ4. How accurately can future municipal debt burden be predicted out of time at one-, two-, and three-year horizons, and how stable are model rankings across thresholds, folds, simple baselines, and tail-sensitive loss behavior?

Figure 1 summarizes the two-track design and shows that the cross-country inferential comparison is secondary rather than a pooled multinational analysis.

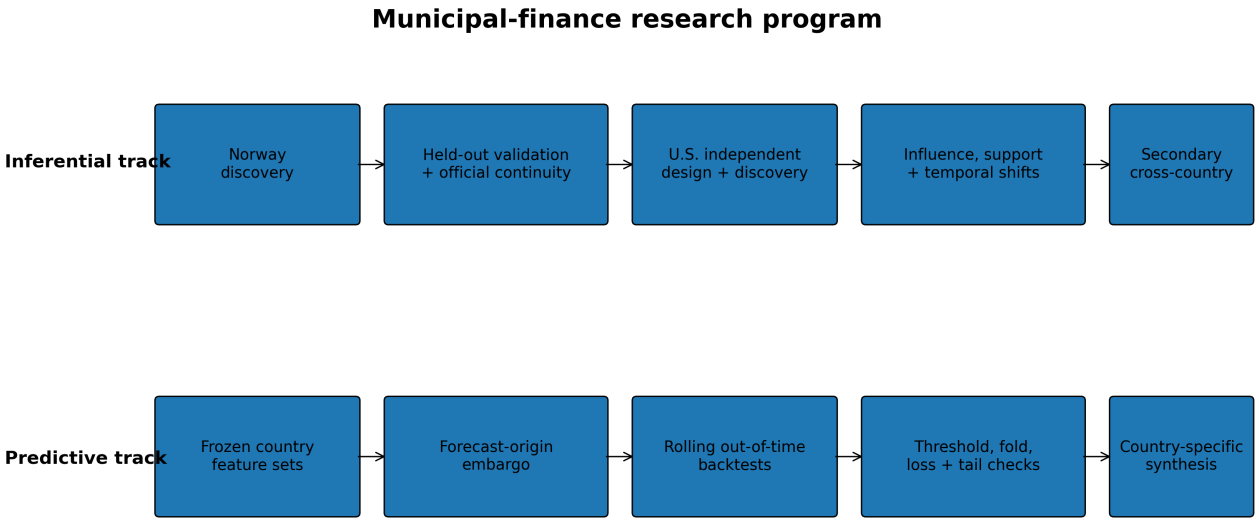


Figure 1: Figure 1. Revised research program. Inference and prediction are maintained as separate evidence layers, and cross-country inference is secondary.

2. Institutional and data context

2.1 Norway

Norwegian municipal data come from Statistics Norway's KOSTRA reporting system (Statistics Norway 2026b). The frozen source set uses KOSTRA tables 12134, 12364, 13502, 12209, 11791, 11789, 07459, 13536, and 11618. Models are municipality-year panels in which financial characteristics measured at feature year t are linked to outcomes at horizons $h = 1, 2, 3$.

Norwegian municipality identifiers are not perfectly stable over the study period. The 2020 reform involved both actual mergers and widespread renumbering, and the 2024 changes included a split of Haram from Ålesund plus additional code changes associated with county restructuring (Kartverket 2026, 2023). The official-continuity analysis therefore uses Statistics Norway's official KLASS municipality-change mappings (Statistics Norway 2026a). One-to-one old-code-to-new-code changes are harmonized as continuity-preserving identities before constructing $t \to t + h$ pairs; non-one-to-one merger or split structures are excluded from the continuity-stable sensitivity panel.

The original discovery target years run through 2021. Held-out temporal validation uses target years 2022-2025. The continuity analysis is a sensitivity layer rather than a replacement of the raw-code discovery program.

2.2 United States

The U.S. analysis uses the Census Bureau State and Local Government Finances program (U.S. Census Bureau 2026b, 2026c, 2026a). The frozen primary universe contains county, municipal, and township governments: 3,030 counties, 19,504 municipalities, and 16,235 townships, for 38,769 governments before analysis-specific completeness restrictions. The study excludes state governments, school districts, and special districts from the primary local-government universe.

The source series spans 2017-2024. U.S. discovery models use target years through 2022, while held-out validation uses target years 2023-2024. Census of Governments years ending in 2 and 7 canvass the full universe, while intervening annual-finance years survey a sample of local governments (U.S. Census Bureau 2026d). The current pipeline does not separately exclude observations by imputation status, so the analysis should be interpreted as applying to the published Census finance files rather than to a complete-case subset defined by an imputation flag. The held-out validation design also requires observations in both 2023 and 2024 for the retained subset, so its estimand is conditional on that longitudinal availability.

The U.S. construct audit identified a 2022 measurement/classification bridge in long-term debt and capital-outlay coding. The primary outcome is gross long-term debt burden. Focal predictors are capital-outlay intensity, own-source revenue share, and interest burden, with log general revenue as the scale control. Cross-country equivalence is not assumed.

2.3 Focal variable definitions

Table 1. Focal variables and operational definitions.

Country	Variable	Operational definition	Unit
Norway	Net loan debt burden	KOSTRA key figure: net loan debt as a percentage of gross operating revenues	%
Norway	Accumulated balance	KOSTRA key figure: annual accumulated surplus/deficit as a percentage of gross operating revenues	%
Norway	Home-service use, age 80+	Share of inhabitants aged 80 years and over using home-based services	%
Norway	Self-investment financing	Self-investment financing as a percentage of total gross investment expenditures	%

Country	Variable	Operational definition	Unit
Norway	Unrestricted revenue per capita	KOSTRA unrestricted revenues per capita	NOK/person
United States	Gross long-term debt burden	100 x gross long-term debt / general revenue	%
United States	Capital-outlay intensity	100 x capital outlay / general revenue	%
United States	Own-source revenue share	100 x own-source general revenue / general revenue	%
United States	Interest burden	100 x interest on general debt / general revenue	%

The U.S. ratio constructs therefore share general revenue as denominator. This common denominator is one reason denominator-tail diagnostics are treated as a substantive robustness check rather than a cosmetic outlier screen.

3. Methods

3.1 Staged design-freeze workflow

The project used a staged design-freeze workflow. Source definitions, construct decisions, and model protocols were frozen before the later stages to which they applied. The project was **not formally preregistered**. Repository history records pre-result freezes, post-diagnostic technical corrections, and later robustness extensions separately. This distinction is important because a Git timestamp documents chronology but does not itself constitute preregistration.

3.2 Norway inferential models

For municipality i and target year $\tau = t + h$, the main Norway specification is

$$Y_{i,\tau} = \alpha_i + \lambda_\tau + \beta X_{i,t} + \gamma' Z_{i,t} + \varepsilon_{i,\tau},$$

where α_i are municipality fixed effects, λ_τ are target-year fixed effects, $X_{i,t}$ is the focal financial predictor, and $Z_{i,t}$ contains frozen controls for total population and age structure. Standard errors are CR1 cluster-robust by municipality (Cameron and Miller 2015). Discovery multiplicity is controlled using Benjamini-Hochberg false-discovery-rate adjustment (Benjamini and Hochberg 1995).

Three Norway focal associations passed discovery FDR control. The continuity and temporal-validation analyses do not reselect those associations; they ask whether the same three remain stable after official municipality continuity harmonization and in held-out years.

3.3 U.S. inferential models and temporal shifts

The U.S. model uses government fixed effects, target-year fixed effects, the three frozen focal predictors, and log general revenue. CR1 standard errors are clustered by government. FDR is controlled within each outcome-by-horizon family of the three focal predictors.

The revision adds two diagnostics. First, the h=2 and h=3 interest-burden coefficients are subjected to exact influential-cluster deletion, trimming, winsorization, and low-denominator exclusion. Second, discovery and validation coefficients are compared in a fully period-specific stacked model. Entity fixed effects and all focal/control slopes are allowed to differ by period, while clustering remains at the original-government level. This stacked specification reproduces the separately fitted discovery and validation slopes to numerical precision and yields the covariance needed for a formal test of $\beta_{validation} - \beta_{discovery}$.

3.4 Cross-country harmonization

Constructs are classified as Class A (accounting-identical), Class B (conceptually harmonizable but not accounting-identical), or Class C (not suitable for direct comparison). The focal cross-country debt/revenue concepts are Class

B. Raw coefficient magnitudes and raw RMSE values are therefore not compared between countries. Missing frozen models are treated as unavailable evidence, not as null effects.

The final cross-country inferential matrix is intentionally small: one common frozen predictor concept across three horizons. A descriptive count such as "2 of 3 signs agree" is not treated as a replication rate because three cells provide too little information for such a percentage to be meaningful.

3.5 Predictive design and forecast-origin correction

Prediction uses current debt burden and a frozen country-specific feature set to forecast future debt burden at $h = 1, 2, 3$. Eight candidate models are evaluated: Historical Mean, Persistence, pooled AR(1), ordinary linear regression, Ridge (Hoerl and Kennard 1970), ElasticNet (Zou and Hastie 2005), Random Forest (Breiman 2001), and histogram gradient boosting (Friedman 2001). Implementations use scikit-learn (Pedregosa et al. 2011).

Evaluation is retrospective rolling out of time (Tashman 2000). The corrected information rule is

Let T_{train} denote the target year of a training observation and Y the test target year. The information rule is

$$T_{\text{train}} \leq Y - h.$$

For a test target year Y and horizon h, the forecast origin is $Y - h$. Training outcomes dated later than $Y - h$ would not have been observable at the forecast origin and are therefore embargoed. The same rule is used inside temporal tuning. Test features also respect the forecast origin: the feature-timing audit verified in all six country-horizon cells that feature year equals $Y - h$ and target year matches the outer test fold, so no feature dated after the forecast origin enters prediction. Because the embargo becomes stricter as h increases, fewer historical target years are eligible for training and fewer outer folds remain at longer horizons. This correction is especially important for $h > 1$.

Historical Mean is the pooled arithmetic mean of admissible training targets. Persistence uses the current feature-year debt burden Y_t. Pooled AR(1) is a linear regression of future debt burden on current debt burden within the admissible training sample. No global imputation occurs before the outer split.

RMSE is the primary loss, with MAE and R^2 secondary. Selection uses the lowest median outer-fold RMSE. If models fall within a frozen near-tie threshold, the simpler ordering is preferred. Robustness analysis recomputes selection at 0%, 1%, and 5%, reports paired fold differences and descriptive fold-bootstrap intervals, and quantifies the share of squared error contributed by the largest 1% of row-level errors. The bootstrap intervals are descriptive because there are few outer folds and folds are not independent replicates.

3.6 Interpretation boundary

All inferential estimates are observational associations. Fixed effects remove time-invariant unit heterogeneity but do not eliminate time-varying confounding, simultaneity, policy feedback, measurement error, or endogenous fiscal responses. Predictive accuracy does not establish causation. Conversely, the absence of FDR support for a single coefficient does not imply that a variable is useless for prediction.

4. Results

4.1 Norway: discovery support weakens under temporal and continuity tests

The original Norway discovery program produced three FDR-supported focal associations. Table 2 shows the original discovery coefficient and q-value together with the official-continuity estimates in discovery and validation.

Table 2. Norway focal inferential results after official municipality-continuity correction.

Association	h	Original discovery β (q)	Stable-ID discovery β [95% CI]; N / clusters	Stable-ID validation β [95% CI]; N / clusters	Revised interpretation
Self-investment → debt burden	1	-0.046294 (0.004757)	-0.058285 [-0.091229, -0.025341]; 1,857 / 324	+0.004119 [-0.036410, +0.044647]; 1,234 / 319	Discovery robust; validation attenuates to zero
Revenue/capita → accumulated balance	2	-0.000074 (0.016148)	-0.000087 [-0.000133, -0.000041]; 1,533 / 321	+0.000017 [-0.000025, +0.000059]; 1,220 / 305	Discovery robust; held-out sign reversal
Self-investment → home-service use (80+)	2	+0.013265 (0.049088)	+0.004672 [-0.003567, +0.012912]; 1,491 / 320	+0.006648 [-0.000715, +0.014010]; 1,149 / 303	Same direction; support fragile

The official change table contained 512 municipality-change rows. Of these, 360 were one-to-one code changes and 152 were non-one-to-one structural changes. Within the study panel, 185 raw municipality codes were excluded as structural-change codes, while one-to-one code histories were collapsed into 339 continuity-stable entity chains. These counts are not additive municipality counts: the 185 figure refers to raw historical codes excluded for structural change, whereas 339 refers to harmonized entity chains retained after code-history collapse. The continuity correction therefore changes the interpretation of the debt result from a same-direction held-out estimate to attenuation around zero; the small positive point estimate is well inside a confidence interval spanning both signs and is not interpreted as evidence of a positive association. The home-service association remains positive across periods, but both continuity-adjusted confidence intervals include zero. The revenue association retains a clear negative discovery estimate and reverses direction in validation.

Direct coefficient-shift tests confirm that the three Norway associations have different temporal behaviors (Table 3). The revenue/balance shift and the self-investment/debt shift remain supported after Benjamini-Hochberg adjustment across the three focal shift tests, whereas the home-service shift does not. The self-investment/debt result should therefore be described as a supported attenuation/change from the discovery association to approximately zero in validation, not as evidence for a positive validation association.

Table 3. Formal Norway discovery-versus-validation coefficient shifts on the official stable-ID panel.

Association	h	Discovery β	Validation β	Shift [95% CI]	p	BH q	N / clusters
Revenue/capita → accumulated balance	2	-0.000087	+0.000017	+0.000105 [0.000039, 0.000171]	0.001921	0.005762	2,753 / 324
Self-investment → debt burden	1	-0.058285	+0.004119	+0.062404 [0.014295, 0.110513]	0.011011	0.016517	3,091 / 338
Self-investment → home-service use (80+)	2	+0.004672	+0.006648	+0.001975 [-0.009320, 0.013270]	0.731797	0.731797	2,640 / 324

The stacked period-specific model reproduces the separate-period slopes to numerical roundoff (maximum absolute differences 2.8×10^{-17} in discovery and 4.3×10^{-18} in validation). The three shift tests form a separate robustness family and do not alter the original discovery FDR family.

4.2 United States: influence sensitivity and formal temporal shifts

At h=1, no focal U.S. predictor passed discovery FDR control. At h=2 and h=3, the untrimmed pooled model produced very large negative interest-burden coefficients: -22.981822 (SE 0.227202) and -44.726126 (SE 0.195013), respectively. These values were not stable to influence and denominator diagnostics.

At h=2, deleting the most influential cluster reduced the coefficient to -1.538, 0.5/99.5 trimming to -2.043, winsorization to -3.591, and the original one-period low-denominator diagnostic to -1.878. The sign remained negative in these influence diagnostics, but the magnitude fell by roughly 84-93% across the directly comparable top-one-cluster deletion, trimming, winsorization, and low-denominator checks. (The broader cumulative top-5/10/20 deletion sequence gives a wider reduction range.) At h=3, deleting the most influential cluster changed the coefficient to +13.454; trimming and winsorization reduced the magnitude to about -1.216 and -1.136; the original one-period low-denominator diagnostic changed it to about +14.213. Table 6 uses a different, stricter rule for temporal comparability: the general-revenue cutoff is frozen from discovery and then applied identically to both discovery and validation. Under that rule the discovery coefficients are -6.076 at h=2 and +14.219 at h=3. The two sets of values therefore answer different sensitivity questions rather than representing a numerical contradiction. The primary h=3 direction is not stable.

Window-specific sensitivity estimates also varied substantially (Table 4), so the pooled long-horizon coefficient should not be interpreted as a stable association across measurement and macro-fiscal regimes.

Table 4. U.S. interest-burden coefficient by selected discovery-period window.

h	Window	Interest-burden beta	SE	Within-R^2
2	Full discovery	-22.982	0.227	0.941
2	Pre-2022 measurement redesign	-16.090	6.447	0.022
2	Pre-pandemic	-9.814	1.650	0.004
2	2021-2022 pandemic/ARPA window	-0.888	1.246	0.009
3	Full discovery	-44.726	0.195	0.972
3	Pre-2022 measurement redesign	+0.057	0.974	0.000
3	2021-2022 pandemic/ARPA window	+128.986	92.440	0.201

The window analysis is descriptive because these windows differ in sample size, measurement regime, and macroeconomic context.

Formal discovery-versus-validation tests quantify the untrimmed change (Table 5), but the robustification analysis shows that the temporal-shift interpretation itself is specification-sensitive.

Table 5. U.S. untrimmed period-specific coefficient shifts.

Predictor	h	Discovery β	Validation β	Shift [95% CI]	N / clusters
Interest burden	2	-22.981822	+6.908757	+29.890579 [23.913436, 35.867722]	35,909 / 8,656
Interest burden	3	-44.726126	-0.551604	+44.174522 [41.636904, 46.712140]	27,272 / 6,324
Own-source revenue share	2	+2.177561	-5.790767	-7.968328 [-14.402701, -1.533955]	35,909 / 8,656

No h=1 focal predictor had a discovery-versus-validation shift confidence interval excluding zero. The own-source-revenue h=2 row is included because the formal shift diagnostic was applied to all frozen focal U.S. predictors, not because own-source revenue was FDR-supported in discovery; it is therefore a secondary temporal-change diagnostic rather than a replicated discovery finding.

The robust period-shift analysis reapplied the period-shift test using discovery-frozen robustification rules, so the same trimming/winsorization thresholds were imposed on discovery and validation. The results were not specification-invariant (Table 6). At h=2, trimming reduced the coefficients to -2.043 and -1.381 and the shift confidence interval

included zero; winsorization retained negative coefficients in both periods (-3.591 and -1.217) and supported only attenuation, not sign reversal. At h=3, trimming produced a small supported sign change (-1.216 to +0.863), whereas winsorization produced no supported shift. Excluding the lowest 1% of discovery general-revenue denominators yielded still different results at both horizons. These robustifications are sensitivity analyses rather than a new multiplicity-controlled family of confirmatory hypothesis tests; no single specification is elevated as "the" finding. The h=3 trimmed shift has a 95% CI lower bound of only 0.086 and is therefore described as marginal rather than strong evidence of change. Thus the large untrimmed period shifts are not robust evidence of a single temporal break.

Table 6. Robust U.S. interest-burden period-shift sensitivity using discovery-frozen rules.

h	Specification	Discovery β	Validation β	Shift [95% CI]	Result
2	Untrimmed	-22.982	+6.909	+29.891 [23.913, 35.868]	Reversal; influence-dominated
2	Trim 0.5/99.5	-2.043	-1.381	+0.661 [-1.208, 2.530]	No supported shift
2	Winsor 0.5/99.5	-3.591	-1.217	+2.374 [0.259, 4.490]	Supported attenuation
2	Exclude lowest 1% revenue	-6.076	+6.929	+13.006 [7.011, 19.000]	Reversal under this rule
3	Untrimmed	-44.726	-0.552	+44.175 [41.637, 46.712]	Attenuation; influence-dominated
3	Trim 0.5/99.5	-1.216	+0.863	+2.079 [0.086, 4.072]	Small supported sign change
3	Winsor 0.5/99.5	-1.136	+0.752	+1.887 [-0.542, 4.316]	No supported shift
3	Exclude lowest 1% revenue	+14.219	-0.063	-14.282 [-43.551, 14.988]	No supported shift

The influence audit also showed that the same finance-key entity (`212069105387`) ranked first at both h=2 and h=3. Its discovery rows include general revenue as low as 22, interest burden as high as 18,181.8%, and future debt burden as high as 555,555.6%. Ratio reconstruction from the underlying numerators and denominators passed, so these are internally consistent extreme observations rather than ratios created by a coding error. Other leading influential entities likewise contain small denominators and/or extreme interest and future-debt ratios. A repository-wide identifier lookup scanned 145 repository CSV/TSV files for the six most influential finance-key IDs. All IDs were present in continuity/audit files, but the frozen repository did not expose human-readable government-name fields. The appendix therefore reports exact finance-key identifiers, government-type codes, and diagnostic characteristics rather than inferring names from an unverified external mapping.

4.3 U.S. sample-composition sensitivity

The held-out 2023-2024 sample is almost entirely nested inside the discovery universe, but it is a substantially smaller retained subset. The retained governments are also systematically larger: the median of government-level median general revenue is about 44,342 versus 6,812 at h=1, 43,121 versus 5,678 at h=2, and 41,547 versus 5,957 at h=3 for retained versus non-retained governments. Retained governments also contribute more discovery observations per entity. A usable government-type field was not exposed in this audit, so no type-composition claim is made. At h=1, 4,210 of 4,213 validation governments also appear in discovery; at h=2, 4,206 of 4,209; and at h=3, all 4,208 validation governments appear in discovery. Validation therefore conditions heavily on governments observed in both held-out years.

Restricting discovery and validation to the same governments changes the interest-burden shift interpretation materially. At h=2, the discovery coefficient shrinks from -22.982 in the full sample to -5.898 on common governments, while validation remains +6.910; the resulting shift remains supported (95% CI [6.238, 19.379]). At h=3, however, the common-government discovery coefficient becomes +22.411 and the shift to -0.552 is no longer supported (95% CI [-69.240, 23.315]). Thus composition is not the sole explanation for h=2 but is an important part of the h=3 apparent temporal change.

The own-source-revenue h=2 reversal also persists on the common-government sample (+2.299 to -5.857; shift CI [-16.195, -0.117]), reinforcing its status as a secondary period-change diagnostic rather than an FDR-supported discovery replication.

These results strengthen the overall interpretation: the large U.S. interest coefficients are influence- and denominator-dominated, and evidence for a temporal break is additionally sensitive to sample composition.

Fixed absolute revenue-floor sensitivity

As an additional post-hoc robustness analysis, the U.S. period-shift models were re-estimated after requiring both feature-year and target-year general revenue to be at least $1 million. The absolute threshold was fixed before this sensitivity run and is not presented as part of the original staged design freeze. The rule retained approximately 92-95% of complete rows.

For interest burden, h=1 showed no supported period shift (-9.781 to -8.529; shift CI [-8.837, 11.341]). At h=2, a smaller sign reversal persisted (-5.018 to +7.830; shift CI [2.629, 23.068]). At h=3, the discovery estimate was unstable (+53.175) and the period shift was not supported (shift CI [-149.886, 43.805]). The h=2 own-source-revenue shift also lost support under this absolute floor.

This sensitivity therefore reinforces a narrower conclusion: the h=2 interest change is the most persistent U.S. period-change signal across alternative support restrictions, but its discovery magnitude is far smaller than the original untrimmed coefficient. The h=3 temporal-change interpretation is not robust.

Table 7. Interest-burden shift under the fixed $1 million general-revenue floor.

h	Discovery β	Validation β	Shift [95% CI]	N / clusters
1	-9.781	-8.529	+1.252 [-8.837, 11.341]	37,688 / 7,885
2	-5.018	+7.830	+12.848 [2.629, 23.068]	33,368 / 7,796
3	+53.175	+0.134	-53.041 [-149.886, 43.805]	25,627 / 5,866

4.4 Cross-country inferential comparison

Only the revenue-structure concept had frozen models suitable for direct Norway-U.S. Class-B comparison across the three horizons. There was no horizon at which both countries established held-out FDR-supported replication. Because the matrix contains only one predictor concept by three horizons, descriptive sign agreement is reported only as context and not as a replication rate.

The inferential conclusion is therefore narrow and secondary to the country-specific analyses: **no statistical Norway-U.S. replication was established**.

Table 8. Secondary Norway-U.S. revenue-structure comparison.

h	Discovery layer	Held-out layer	Joint interpretation
1	Same direction; no dual-country FDR support	Directions diverge	No replication
2	Same direction; no dual-country FDR support	Same direction, but both country patterns change over time and lack dual held-out FDR support	No statistical replication
3	Directions diverge	Directions diverge	No replication

The table reports direction and FDR-support status rather than raw coefficients, because the measures are Class-B harmonizations rather than accounting-identical constructs and exact q-values are not treated as cross-country effect comparators. The evidence instead shows that apparently strong within-country associations can fail through sign reversal, magnitude collapse, influence sensitivity, or institutional/measurement change.

4.5 Forecast-origin-corrected predictive results

Table 9 reports the corrected predictive evaluation depth. The Norway h=1 count of 780 refers to raw municipality codes appearing across the full 2019-2025 prediction window, not 780 simultaneously existing municipalities. Official continuity mapping reduces those raw histories to 321 nonstructural stable IDs plus 170 structural raw-code histories; multiple historical codes can collapse into the same stable entity. At h=2 and h=3, the later windows contain 356 raw codes, corresponding to 305 nonstructural stable IDs plus 51 structural raw-code histories.

Table 9. Predictive sample sizes.

Country	h	Evaluated prediction rows	Raw unique IDs	Stable nonstructural IDs	Outer folds	Target years
Norway	1	2,201	780	321	7	2019-2025
Norway	2	1,421	356	305	6	2020-2025
Norway	3	1,065	356	305	5	2021-2025
United States	1	23,121	6,336	n/a	4	2021-2024
United States	2	19,125	8,642	n/a	3	2022-2024
United States	3	10,499	6,288	n/a	2	2023-2024

For Norway, "raw unique IDs" counts municipality codes appearing anywhere in the multi-year backtest; the stable-ID column reports nonstructural entity chains after official one-to-one code harmonization. Structural raw-code histories are reported separately in the supplement and are not additive to the stable-ID count.

Under the 1% simplicity rule, Norway selects Persistence at all three horizons. At h=1, however, ElasticNet is the strict numerical median-RMSE winner (10.6239 versus 10.6784 for Persistence), a gap of only 0.51%; selection therefore depends on the simplicity rule. At h=2 and h=3, Persistence is the direct numerical winner with median RMSE 15.1972 and 18.7731. Pooled AR(1) has median RMSE 16.0319 and 20.1701 at those horizons.

Norway's stronger result is relative to the pooled Historical Mean baseline: Persistence wins all 7/7, 6/6, and 5/5 outer folds at h=1, h=2, and h=3, respectively. Against pooled AR(1), however, Persistence wins only 4/7, 3/6, and 3/5 folds, and all descriptive bootstrap intervals for the mean paired RMSE difference cross zero. This is not inconsistent with Persistence having the lower median RMSE at h=2 and h=3: model selection summarizes the median across folds, whereas the win count records how many individual folds favor each model. The evidence therefore supports state-based predictability, not universal superiority of the Persistence rule.

The U.S. 1% selection pattern is Persistence / Historical Mean / Persistence across h=1/2/3. The principal instability is not a wholesale change of architecture at every horizon but weak separation and threshold sensitivity: at h=3 the 0% and 1% rules select Persistence, whereas at a 5% near-tie threshold all eight candidate models enter the near-tie set and the simplicity order selects Historical Mean instead. Fold-level comparisons are also weak: the selected model wins only 2/4, 2/3, and 1/2 folds against the Historical Mean or Persistence reference as applicable, and paired-fold bootstrap intervals cross zero.

Table 10. Model-selection sensitivity to the near-tie threshold.

Country	h	0% rule	1% rule	5% rule	Near-tie count at 5%
Norway	1	ElasticNet	Persistence	Persistence	6
Norway	2	Persistence	Persistence	Persistence	1
Norway	3	Persistence	Persistence	Persistence	1
United States	1	Persistence	Persistence	Persistence	1
United States	2	Historical Mean	Historical Mean	Historical Mean	1
United States	3	Persistence	Persistence	Historical Mean	8

The most consequential U.S. diagnostic is error concentration. Recomputed directly from the frozen common-support prediction rows, the largest 1% of row-level prediction errors contribute a median 99.81%, 99.32%, and 99.89% of total squared error at h=1, h=2, and h=3. Removing those observations reduces RMSE by 95.89%, 91.73%, and 97.06%. Norway's corresponding shares are 21.20%, 23.07%, and 21.73%, with RMSE reductions around 11% (Figure 2).

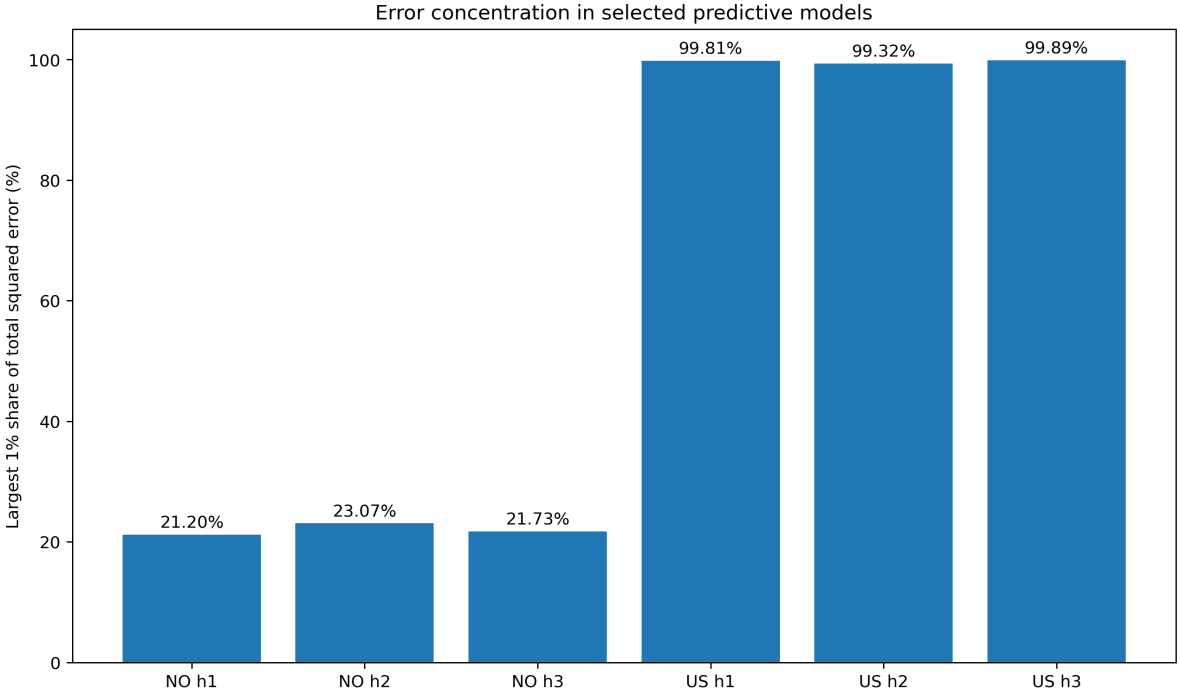


Figure 2: Figure 2. Share of squared error contributed by the largest 1% of row-level errors in the selected predictive models.

The U.S. h=2 result also illustrates loss-function dependence: Historical Mean is selected by RMSE, while its summary MAE ratio relative to Persistence is about 1.20. A model that better avoids a small number of catastrophic squared errors can therefore rank better under RMSE while being worse for the typical absolute error.

Table 11. Secondary robust-loss diagnostics for the selected predictive models.

Country	h	1% selected model	Median top-1% share of squared error	Median RMSE reduction after removing top 1%	MAE ratio vs Persistence
Norway	1	Persistence	21.20%	10.73%	1.00
Norway	2	Persistence	23.07%	11.81%	1.00
Norway	3	Persistence	21.73%	10.97%	1.00
United States	1	Persistence	99.81%	95.89%	1.00
United States	2	Historical Mean	99.32%	91.73%	1.20
United States	3	Persistence	99.89%	97.06%	1.00

These are labeled sensitivity diagnostics rather than a replacement primary loss. They show that U.S. RMSE rankings primarily describe performance on extreme rows and are not representative of typical governments. The final reporting reconciliation recomputed every selected-model median RMSE directly from the frozen row-level predictions and matched the frozen predictive-benchmark metrics exactly (maximum absolute difference 0). The reconciled U.S. h=2 Historical Mean median RMSE is 1328.318 on the common evaluation support.

All-model MAE rankings show that the loss-function dependence is broader than the selected-model comparison (Table 12). On common support, Norway selects Persistence by median MAE at every horizon. In the United States, MAE selects Persistence at h=1, Histogram Gradient Boosting at h=2, and Random Forest at h=3, whereas RMSE selects

Persistence, Historical Mean, and Persistence. Under the fixed $1 million floor, U.S. MAE winners are Persistence, Persistence, and Random Forest, while RMSE winners are pooled AR(1), Historical Mean, and pooled AR(1).

Table 12. Median-MAE winner versus median-RMSE winner.

Sample	h	MAE winner	RMSE winner
Norway common support	1	Persistence	ElasticNet
Norway common support	2	Persistence	Persistence
Norway common support	3	Persistence	Persistence
U.S. common support	1	Persistence	Persistence
U.S. common support	2	Histogram Gradient Boosting	Historical Mean
U.S. common support	3	Random Forest	Persistence
U.S. fixed $1M floor	1	Persistence	pooled AR(1)
U.S. fixed $1M floor	2	Persistence	Historical Mean
U.S. fixed $1M floor	3	Random Forest	pooled AR(1)

The complete eight-model MAE and RMSE rankings are reported in the supplement.

5. Discussion

5.1 Temporal generalizability is not captured by discovery significance

The Norway results show why discovery support and temporal generalizability should be separated. All three focal associations passed discovery FDR control, but the official continuity and held-out analyses produce different temporal behaviors. Formal shift tests support change for the revenue/balance and self-investment/debt associations after BH adjustment, while the home-service association does not show a supported shift. These outcomes are not well summarized by comparing whether each period is individually significant.

The untrimmed U.S. coefficient-shift tests initially suggested temporal reversal or attenuation, but the robustification analysis shows that the shift is specification-sensitive: trimming removes the h=2 shift, winsorization removes the h=3 shift, and low-denominator exclusion produces still another pattern. The common-government analysis further shows that h=3 lacks a supported shift on the retained common sample, while h=2 retains a smaller sign reversal. The appropriate interpretation is therefore influence/denominator dominance with horizon-specific, sample-composition-sensitive temporal evidence; the relevant evidence is the pattern across untrimmed, robustified, and common-government analyses rather than one specification or a comparison of p-values across periods.

5.2 Extreme coefficients can be properties of the sample and measurement regime

The original U.S. interest results appeared exceptionally precise, but the influence audit shows that this precision was driven by a small number of extreme clusters and denominator tails. Removing those observations reduced the h=2 magnitude by most of its value and made the h=3 direction unstable, so the headline result is first an influence problem and only secondarily a temporal-generalizability question. This does not prove that interest exposure is irrelevant to local fiscal dynamics. It shows that the specific pooled coefficients cannot be interpreted as stable population-wide structural parameters. The distinction matters in municipal-finance datasets, where ratios can contain small denominators, fiscal crises are heavy-tailed, and accounting/classification changes can coincide with macroeconomic shocks.

5.3 Predictability and inferential association answer different questions

The predictive results do not validate or invalidate the inferential coefficients. Norway's debt burden is strongly predictable relative to a historical mean because current debt burden contains substantial state information. Yet that does not identify a causal effect of debt on future debt. In the United States, large inferential interest coefficients

coexist with unstable predictive rankings and extreme tail dependence. These observations are compatible because inference and prediction optimize different estimands and loss functions.

The forecast-origin correction is central to this conclusion. A multi-year backtest that permits training on outcomes not yet observed at the forecast origin can overstate available information. After enforcing $T_{\text{train}} \leq Y - h$, the old U.S. h=3 Random Forest selection is superseded by Persistence under the 1% rule, and h=2 remains Historical Mean. The correction changes model rankings without changing the underlying source panel.

5.4 The Norway-U.S. comparison is descriptive, not a claim of institutional equivalence

The two national systems differ in accounting, intergovernmental finance, municipal structure, and data-production rules. The Class-B harmonization rule is therefore deliberately conservative. The study does not pool coefficients across countries and does not compare raw RMSE magnitudes. The appropriate conclusion is that no common inferential replication or common predictive architecture was established under the frozen designs.

Norway exhibits comparatively stable state-based predictability and limited tail dominance. U.S. prediction is much more tail-sensitive, and its model hierarchy is weakly separated and threshold-sensitive at the three-year horizon. These differences may reflect institutional structure, measurement design, fiscal heterogeneity, or sample composition, but the current study does not identify which mechanism is causal.

5.5 What the staged design-freeze workflow contributes

The project was not preregistered, and the paper does not present it as such. Its methodological contribution is instead an auditable separation of stages: source and construct decisions were frozen before later tests; post-diagnostic corrections were logged; and later robustness corrections were added without rewriting earlier outputs as if they had always been planned. This is a weaker claim than preregistration but a stronger and more transparent claim than an undocumented exploratory workflow.

6. Limitations

First, all inferential models are observational. Municipality or government fixed effects do not remove time-varying confounding, endogenous policy responses, or reverse feedback. Second, U.S. annual finance data combine census years with sampled intervening years and published-data processing; the current analysis does not separately condition on imputation status. Held-out validation also requires complete 2023 and 2024 rows. The common-government analysis shows that validation governments are almost entirely contained in discovery but form a much smaller retained subset that is substantially larger by general revenue; common-government shift tests therefore accompany the primary period comparison so that coefficient change is not automatically attributed to time when sample composition also changes. Third, the U.S. 2022 classification bridge coincides with major fiscal-regime changes associated with the pandemic and federal transfers, so measurement and regime effects cannot be fully disentangled.

Fourth, Norway's official municipality-continuity sensitivity excludes non-one-to-one structural changes rather than reconstructing counterfactual pre/post merger entities. Fifth, the cross-country comparison is sparse because only constructs with frozen models in both countries are eligible; absence of a model is not evidence of no effect. Sixth, predictive fold counts become small at longer horizons, especially the U.S. h=3 design with two folds, so fold-bootstrap intervals are descriptive rather than inferential confidence intervals in a repeated-sampling sense.

Finally, U.S. RMSE is dominated by extreme observations. That is itself a substantive result for squared-error forecasting, but it means the selected model should not be described as the universally best model for a typical government. MAE and fixed-support analyses are therefore reported only as secondary sensitivities rather than replacing the frozen RMSE primary loss.

7. Data, code, and reproducibility record

Norway source data are from Statistics Norway/KOSTRA and the official KLASS municipality classification API (Statistics Norway 2026b, 2026a). U.S. source data are from the Census Bureau State and Local Government Finances program and its technical documentation (U.S. Census Bureau 2026b, 2026c, 2026a). Raw third-party source files are not redistributed in the publication package; reconstruction instructions and source identifiers are provided instead.

The repository records the staged analysis history, including the observed-year universe correction, U.S. common-support correction, forecast-origin correction, U.S. influence audit, official Norway continuity analysis, and predictive robustness analyses. The reproducibility package accompanying this manuscript contains the staged analysis scripts, correction logs, source identifiers, and release manifest. A repository URL and archival DOI are intentionally omitted from this manuscript version until an actual public release is created; they must be inserted from the real release record rather than inferred or preassigned.

8. Declarations

Ethics. The study analyzes public government-finance and municipal administrative statistics and does not involve direct recruitment, intervention, or private individual-level participant data.

Author contributions. Abdulrazzag Al Alewy conceived the study, developed the analysis program, performed the computational analyses, interpreted the results, and prepared the manuscript.

AI-assisted work. Generative AI tools were used for coding assistance, drafting, editing, and documentation support. The author remains responsible for the study design, execution, verification of outputs, interpretation, citations, and final manuscript.

9. Conclusion

The revised evidence does not support a universal municipal-finance relationship that replicates cleanly across time and national systems. Norway's discovery associations weaken in distinct ways under official continuity and held-out validation. In the United States, the most striking interest-burden coefficients are dominated by a small set of internally consistent extreme observations and denominator tails. Robustification, common-government restriction, and a fixed $1M revenue floor all sharply reduce the discovery magnitude. Only the h=2 interest shift persists across several alternative support rules; h=3 period change does not. Cross-country inferential evidence therefore remains descriptive rather than replicative.

Prediction produces a different form of instability. Norway shows strong state-based predictability relative to a pooled mean, but Persistence is not demonstrably superior to pooled AR(1) across folds. U.S. model rankings are weakly separated, threshold-sensitive at h=3, loss-function dependent, and heavily controlled by extreme squared errors even on the common evaluation support. The appropriate conclusion is not that one model or one fiscal variable is universally correct, but that temporal, measurement, sample, and loss-function dependence are central features of municipal-finance evidence.

Appendix A. Frozen predictive hyperparameter grids

The benchmark used deliberately compact grids rather than unrestricted optimization.

Model	Frozen configurations
Ridge	alpha in {0.1, 1, 10, 100}
ElasticNet	alpha in {0.001, 0.01, 0.1} crossed with l1_ratio in {0.2, 0.5, 0.8}
Random Forest	(300 trees, depth=None, min leaf=1, max_features=sqrt); (300, depth=12, min leaf=2, max_features=0.7); (300, depth=8, min leaf=5, max_features=1.0)
Histogram Gradient Boosting	(learning rate=.05, 200 iterations, 31 leaves, min leaf=20, L2=0); (.10, 150, 31, 20, 0); (.05, 250, 15, 20, 1)
Linear Regression	no hyperparameter tuning
Historical Mean	pooled arithmetic mean of admissible training targets
Persistence	current feature-year debt burden
Pooled AR(1)	pooled linear regression of future debt burden on current debt burden

Linear/Ridge/ElasticNet use training-fold median imputation and training-fold standardization. Random Forest and Histogram Gradient Boosting use training-fold median imputation without scaling. No preprocessing is fit globally before temporal splitting.

Appendix B. Reporting reconciliation

The reporting reconciliation closed two reporting inconsistencies without refitting any model.

1. The U.S. Benjamini-Hochberg family was reconstructed directly from the frozen discovery results using `variable_role == FOCAL_PREDICTOR`. It contains exactly three tests at every horizon: capital-outlay intensity, own-source revenue share, and interest burden. `log_general_revenue_t` is a scale control and is excluded from the FDR family.
2. Secondary predictive-loss metrics were recomputed directly from the frozen row-level prediction files on the same common-support rows used for the frozen predictive-benchmark comparison. Recomputed selected-model median RMSE values matched the frozen predictive-benchmark metrics exactly at all six country-horizon cells (maximum absolute difference = 0). The corrected U.S. h=2 Historical Mean median RMSE is 1328.318, with a median top-1%-SSE share of 99.323% and a median RMSE reduction of 91.733% after removing the top 1% of row-level squared errors.

Appendix C. Influential U.S. finance-key identifiers

The top influential entities from the influence audit were:

Rank context	Finance-key entity	Government type code	Repository name resolution
h=2/h=3 top influence	212069105387	2	ID found; no human-readable name field
h=2 top influence	202147208128	2	ID found; no human-readable name field
h=2/h=3 top influence	211041105321	1	ID found; no human-readable name field
h=2/h=3 top influence	182153207971	2	ID found; no human-readable name field
h=2/h=3 top influence	212235165990	2	ID found; no human-readable name field
h=3 top influence	212111208165	2	ID found; no human-readable name field

For these entities, the numerator/denominator reconstruction checks passed. The identifiers are therefore reported exactly as present in the frozen analysis repository. No external name mapping was introduced after the fact.

Appendix D. Formal Norway temporal-shift tests

The three focal discovery associations were tested as a separate robustness family using stable municipality identities, entity-by-period fixed effects, target-year fixed effects, and CR1 clustering by stable municipality. Separate discovery/validation slopes from the stacked model match separate-period fits to numerical roundoff.

Association	h	Shift	95% CI	p	BH q
Revenue/capita → accumulated balance	2	+0.000105	[0.000039, 0.000171]	0.001921	0.005762
Self-investment → debt burden	1	+0.062404	[0.014295, 0.110513]	0.011011	0.016517
Self-investment → home-service use (80+)	2	+0.001975	[-0.009320, 0.013270]	0.731797	0.731797

Appendix E. Fixed absolute revenue-floor and all-model MAE sensitivity

The post-hoc absolute support rule requires feature-year and target-year general revenue to be at least $1 million.

Interest-burden period shifts:

h	Discovery beta	Validation beta	Shift 95% CI	Supported?
1	-9.781	-8.529	[-8.837, 11.341]	No
2	-5.018	+7.830	[2.629, 23.068]	Yes
3	+53.175	+0.134	[-149.886, 43.805]	No

The rule retains approximately 92-95% of complete U.S. rows. Its principal implication is that h=2 remains the only long-horizon interest shift that persists under this interpretable absolute floor.

All-model common-support MAE winners: - Norway: Persistence at h=1, h=2, and h=3. - United States: Persistence at h=1, Histogram Gradient Boosting at h=2, and Random Forest at h=3.

All-model fixed-$1M U.S. MAE winners: - h=1 Persistence; - h=2 Persistence; - h=3 Random Forest.

These rankings differ from the RMSE winners and confirm material loss-function dependence.

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